

### 10.1.2

In the graph below, determine whether the following walks are trails, paths, closed walks, circuits, simple circuits, or just walks.

a.  $v_1e_2v_2e_3v_3e_4v_4e_5v_2e_2v_1e_1v_0$

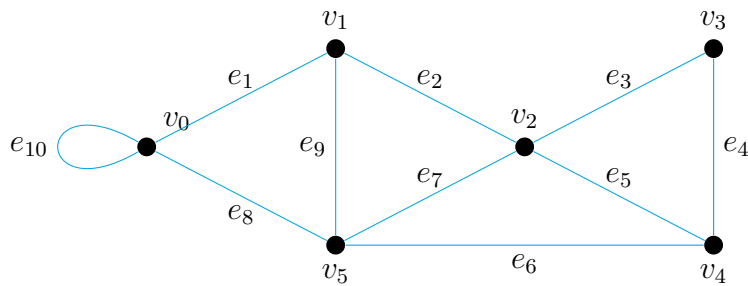
b.  $v_2v_3v_4v_5v_2$

c.  $v_4v_2v_3v_4v_5v_2v_4$

d.  $v_2v_1v_5v_2v_3v_4v_2$

e.  $v_0v_5v_2v_3v_4v_2v_1$

f.  $v_5v_4v_2v_1$



#### Definitions

**Walk** = valid sequence of adjacent vertices/edges

**Trail** = no repeated edges

**Path** = no repeated vertices

**Closed walk** = start = end

**Circuit** = no repeated edges AND start = end

**Simple circuit** = circuit with no repeated vertices except start = end

a.  $v_1e_2v_2e_3v_3e_4v_4e_5v_2e_2v_1e_1v_0$

Edges:  $e_2, e_3, e_4, e_5, e_2, e_1$   
 Repeated edges:  $e_2$   
 Vertices:  $v_1, v_2, v_3, v_4, v_2, v_1, v_0$   
 Repeated vertices:  $v_1, v_2$   
 Start:  $v_1$   
 End:  $v_0$

Repeated edge  $e_2 \rightarrow$  **not a trail**  
 Repeated vertices  $\rightarrow$  **not a path**  
 $v_1 \neq v_0 \rightarrow$  **not a closed walk**  
 Not a closed trail  $\rightarrow$  **not a circuit**  
 Therefore also **not a simple circuit**

**Solution: a.**  $v_1 e_2 v_2 e_3 v_3 e_4 v_4 e_5 v_2 e_2 v_1 e_1 v_0$  is just a walk.

**b.**  $v_2 v_3 v_4 v_5 v_2$

Edges:  $e_3, e_4, e_6, e_7$   
 Repeated edges: No repeated edges  
 Vertices:  $v_2, v_3, v_4, v_5, v_2$   
 Repeated vertices:  $v_2$   
 Start:  $v_2$   
 End:  $v_2$

No repeated edges  $\rightarrow$  **is a trail**  
 Repeated vertices  $v_2 \rightarrow$  **not a path**  
 $v_2 = v_2 \rightarrow$  **is a closed walk**  
 Is a closed trail  $\rightarrow$  **is a circuit**  
 Therefore also **a simple circuit**

**Solution: b.**  $v_2 v_3 v_4 v_5 v_2$  is a simple circuit.

**c.**  $v_4 v_2 v_3 v_4 v_5 v_2 v_4$

|                    |                                     |
|--------------------|-------------------------------------|
| Edges:             | $e_5, e_3, e_4, e_6, e_7, e_5$      |
| Repeated edges:    | $e_5$                               |
| Vertices:          | $v_4, v_2, v_3, v_4, v_5, v_2, v_4$ |
| Repeated vertices: | $v_2, v_4$                          |
| Start:             | $v_4$                               |
| End:               | $v_4$                               |

Repeated edges  $e_5 \rightarrow$  **not a trail**  
 Repeated vertices  $v_2, v_4 \rightarrow$  **not a path**  
 $v_4 = v_4 \rightarrow$  **is a closed walk**  
 Not a closed trail  $\rightarrow$  **not a circuit**  
 Therefore also **not a simple circuit**

**Solution: c.**  $v_4v_2v_3v_4v_5v_2v_4$  is a closed walk.

**d.**  $v_2v_1v_5v_2v_3v_4v_2$

|                    |                                     |
|--------------------|-------------------------------------|
| Edges:             | $e_2, e_9, e_7, e_3, e_4, e_5$      |
| Repeated edges:    | No repeated edges                   |
| Vertices:          | $v_2, v_1, v_5, v_2, v_3, v_4, v_2$ |
| Repeated vertices: | $v_2$                               |
| Start:             | $v_2$                               |
| End:               | $v_2$                               |

No repeated edges  $\rightarrow$  **is a trail**  
 Repeated vertices  $v_2 \rightarrow$  **not a path**  
 $v_2 = v_2 \rightarrow$  **is a closed walk**  
 Is a closed trail  $\rightarrow$  **is a circuit**  
 But is **not a simple circuit**

**Solution: d.**  $v_2v_1v_5v_2v_3v_4v_2$  is a circuit.

**e.**  $v_0v_5v_2v_3v_4v_2v_1$

|                    |                                     |
|--------------------|-------------------------------------|
| Edges:             | $e_8, e_7, e_3, e_4, e_5, e_2$      |
| Repeated edges:    | No repeated edges                   |
| Vertices:          | $v_0, v_5, v_2, v_3, v_4, v_2, v_1$ |
| Repeated vertices: | $v_2$                               |
| Start:             | $v_0$                               |
| End:               | $v_1$                               |

No repeated edges  $\rightarrow$  **is a trail**  
 Repeated vertices  $v_2 \rightarrow$  **not a path**  
 $v_0 \neq v_1 \rightarrow$  **not a closed walk**  
 Not a closed trail  $\rightarrow$  **not a circuit**  
 Therefore also **not a simple circuit**

**Solution: e.**  $v_0v_5v_2v_3v_4v_2v_1$  is a trail.

**f.**  $v_5v_4v_2v_1$

|                    |                      |
|--------------------|----------------------|
| Edges:             | $e_6, e_5, e_2$      |
| Repeated edges:    | No repeated edges    |
| Vertices:          | $v_5, v_4, v_2, v_1$ |
| Repeated vertices: | No repeated vertices |
| Start:             | $v_5$                |
| End:               | $v_1$                |

No repeated edges  $\rightarrow$  **is a trail**  
 No repeated vertices  $\rightarrow$  **is a path**  
 $v_5 \neq v_1 \rightarrow$  **not a closed walk**  
 Not a closed trail  $\rightarrow$  **not a circuit**  
 Therefore also **not a simple circuit**

**Solution: f.**  $v_5v_4v_2v_1$  is a trail and a path.